

南開大學
Nankai University

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College of Cryptology and Cyber Science

Natural Language Processing Course Project · Proposal

A Knowledge-Centric Survey of Hallucination in Large Language Models

Project Proposal · Track A: Survey · Team 8

Pai Peng (2311688) Haotong Li (2312711)

Grade: Class of 2023

Major: Information Security & Law (Double Degree)

Instructor: Qingkai Zeng

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A Knowledge-Centric Survey of Hallucination in Large Language Models

Project Proposal · NLP2026 Course Project, Track A (Survey) · Team XX

1 Team Members & Responsibilities

Each member owns half of the survey and its core papers, so contributions stay traceable.

Member	Responsibilities	Core papers
李浩桐 (2312711)	Types, causes and detection (RQ1–2, detection half of RQ3); runs the shared reading log (one fixed-format card per paper).	[4, 7–9, 15, 16]
澎湃 (2311688)	Mitigation and evaluation (mitigation half of RQ3, RQ4); owns the comparison matrix (Table 1) and the reference check.	[11–13, 20–22]
Both	Anchor surveys; weekly cross-review of cards; joint knowledge-state analysis, open problems and slides.	[1, 2]

2 Background & Motivation

Large language models (LLMs) now answer questions, write code and draft documents for millions of people, yet they regularly produce fluent statements that are false or unsupported: *hallucinations* [1, 2]. The cost is real. In *Mata v. Avianca*, a U.S. court fined lawyers $5,000 over a brief citing cases invented by ChatGPT, which had assured one of them that the cases were real [29]; in 2025, Deloitte agreed to partially refund the Australian government for a report with fabricated references [30]. In ChatGPT-written biographies, only 58% of atomic facts are supported by Wikipedia [13]. Scale alone will not fix this: hallucinations arise from ordinary statistical pressure in pre-training and persist because most benchmarks score "I don't know" as wrong, which rewards guessing [4], and no computable LLM can avoid them entirely [5]. The realistic goal is to detect and reduce them, and to know which tool to use when.

Existing surveys organize methods mainly by technique and pipeline stage [1, 2], and list "knowledge boundaries" as an open direction [1]. Reading them, we noticed that methods differ above all in *where their evidence of truth comes from*: external documents, the model's own repeated answers, or its internal states. This bounds what each can fix. If a model never learned a fact, steering its activations cannot recover it; if it knows the fact but loses it mid-generation, a web search may be overkill. Simhi et al. [11] separate hallucinations caused by missing knowledge from those despite knowledge; we add wrongly learned knowledge as a third case and compare methods along this *knowledge-state* axis.

3 Problem Definition & Research Questions

3.1 Scope and Definitions

In scope: text-only LLMs in open-ended generation (closed-book and retrieval-augmented QA, long-form); factuality, plus faithfulness to retrieved context; detection, mitigation and evaluation methods, 2020–2026. **Out of scope:** vision-language models; pre-LLM task-specific generation, e.g. summarization (background only); agents and tool use (future work). Following [1], a *hallucination* conflicts with verifiable world knowledge (*factuality*) or with the user's input, instructions or the model's earlier output (*faithfulness*). For a fact f a query needs, the model's knowledge, measurable by re-sampling the bare question [11, 12], is:

- **K1 Missing**: f is not in the parameters (long-tail or post-cutoff facts [6, 7]); the share of facts seen only once in pre-training roughly lower-bounds a base model's hallucination rate [4].
- **K2 Wrong**: it holds a false version, e.g. popular misconceptions imitated from human text [8].
- **K3 Unused**: it holds f but fails to use it (sampling noise, snowballing [9], sycophancy [10]).

3.2 Research Questions

RQ1 What? Which taxonomy suits LLMs, and why did the field move from intrinsic/extrinsic [2, 3] to factuality/faithfulness [1]? *Our current answer:* LLMs often answer without any source document, so "contradicts the source" misses the main failure.

RQ2 Why? Which causes in data, training and inference lead to missing, wrong and unused knowledge, and can hallucination be eliminated at all [4, 5]?

RQ3 How? How do detection and mitigation methods differ in truth signal, required access (API vs. open weights), cost, and the knowledge state they address?

RQ4 How well? Do benchmarks separate a wrong answer from an honest "I don't know" [4, 27], and what remains open?

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3.3 Preliminary Taxonomy

Figure 1 is our first taxonomy (built from [1,2] and early reading; to be revised for the mid-term). Its new element is the **K1**–**K3** tags: the knowledge state each cause tends to produce.

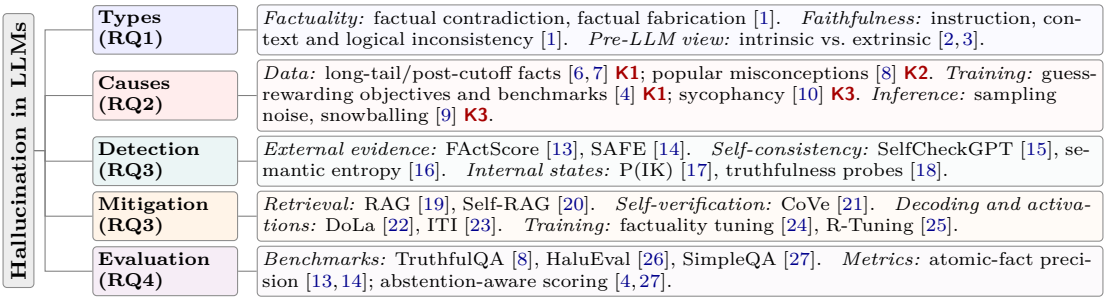


Figure 1. Taxonomy v1. K tags link causes to knowledge states; Table 1 links methods to them.

3.4 Comparison Lens: Which Method Fits Which Knowledge State?

Table 1. Working hypotheses to be tested, grouped by where the truth signal comes from.

Truth signal	Detect / mitigate (examples)	Access; extra cost	K1	K2	K3
External evidence	FActScore, SAFE / RAG, Self-RAG	API; retrieval	✓	✓*	✓
Self-consistency	SelfCheckGPT, sem. entropy / CoVe	API; $k\times$ sampling	○	✗	✓
Internal states	P(IK), truthfulness probes / DoLa, ITI	open weights	○	?	✓
Training signal	— / factuality tuning, R-Tuning	weights; training	○	?	✓

✓ fixes; ○ only detects or abstains; ✗ blind; ? open. *Only if the model trusts the evidence over its memory.

Some cells already have support. Semantic entropy targets arbitrary "confabulations" and explicitly sets apart errors made *consistently* because of erroneous training data [16], i.e. **K2**. GPT-4 recognizes 87% of its own snowballed false claims when asked separately [9], the premise of CoVe [21]. Retrieval helps most on long-tail facts while unaided models stay competitive on popular ones [7], motivating adaptive retrieval [7,20]. If the table holds, only external evidence can *repair* missing or wrong knowledge; methods consulting the model itself can at best flag a gap and abstain. One case puzzles us: activation steering raises Alpaca's TruthfulQA truthfulness from 32.5% to 65.1% [23], so some "wrong" knowledge may really be unused. We will pursue where the **K2**/**K3** boundary lies.

4 Expected Outcomes

1. **An 8-page ACL survey** ($\geq$20 works; 14 core papers† read in depth): taxonomy → causes → detection & mitigation → evaluation → knowledge-state lens → open problems.
2. **Taxonomy v2** (Fig. 1) for the mid-term, with a one-line reason per changed boundary.
3. **A comparison table** of ∼15 methods on fixed axes (truth signal, access, cost, training, knowledge state, benchmark, limitation); each Table 1 cell confirmed, revised or left open.
4. **A "fault-to-fix" guide**: given the model access available and the suspected knowledge state, which method family to try first.
5. **Open problems**: abstention-aware scoring [4,27], long-form factuality [13,14], and whether reasoning models hallucinate less (on PersonQA, o3 33% vs. o1 16% [28]).

5 Project Timeline

Week	Dates	Tasks	Milestone
1	Oct 12–18	Set up Zotero and the card template; read [1,2]	Proposal
2–4	Oct 19–Nov 8	Close-read 6 core papers each, one card per paper	14 cards
5–7	Nov 9–29	Revise taxonomy; fill Table 1; extend list to $\geq$25 works	Mid-term check
8–9	Nov 30–Dec 13	Write the 8-page survey; cross-check all citations	Full draft
10–11	Dec 14–27	Polish the report; make slides; rehearse talk and Q&A	Final report
12	Dec 28–30	Slides due 23:59 the day before; talk on Dec 30	Final talk

Mid-term and report dates are TBA; if the mid-term is early, weeks 5–7 move up and drafting overlaps.

Use of AI tools. We used Claude (Anthropic) to brainstorm the topic, explain concepts, and draft and polish this proposal. Every reference was verified against its original publication; we are responsible for all content.

DRAFT

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Preliminary literature list. † marks the 14 core papers for close reading; entries are grouped by the taxonomy in Fig. 1.

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